[METRIC]

DOD-B-51470(EA)
9 October 1980

MILITARY SPECIFICATION

N-BUTYL MERCAPTAN, TECHNICAL (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of n-butyl mercaptan.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Assay. n-Butyl mercaptan shall contain no less than 97.0 percent by weight n-butyl mercaptan when tested as specified in 4.2.4.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the n-butyl mercaptan produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table I. A representative specimen of approximately 5 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. Sample containers shall be examined for the following defects grouped collectively using an AQL of 4.0 percent defective:

(a) Container damaged or leaking
(b) Palletization incorrect (when required)
(c) Marking incorrect, missing, or illegible

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

TABLE I. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
over 20,000	20

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 10 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Assay.

(a) Apparatus. Use a gas chromatograph equipped with a thermistor or filament detector cell and a liquid sample injection system. The thermal conductivity cell must have a sensitivity which will produce at its highest sensitivity setting at least 90 millimeter (mm) deflection for 15 microliters of a sample containing 1 percent by weight n-butyl mercaptan using the conditions specified in table II. Use a recorder with good stability in the range of 1 to 5 millivolts (mv) with a 2-second pen speed and a chart speed of at least 50 mm per minute.

(b) Column preparation. Prepare two columns containing tricresyl phosphate on 35 to 80 mesh chromosorb by weighing 60 g of the chromosorb into a 250-milliliter (ml) screw-cap bottle, adding 12 g of tricresyl phosphate, and shaking vigorously until the chromosorb appears completely mixed. Pour the contents of the bottle into two lengths of clean aluminum tubing, 6.1 meters (m) and 0.46 m long. Vibrate the columns during the filling operation to insure uniform packing. Plug the ends of the tubing with glass wool and shape the tubing so that it may be conveniently mounted in the chromatograph. Prepare a squalene column in the same manner except use 6 g of chromosorb and 1.2 g of squalene and pour the mixture into a 0.61-m stainless steel column.

(c) Column conditioning. Condition the columns by passing 15 microliters of the material to be analyzed through the column. Subsequent analysis, if made within 30 minutes, can be made with excellent precision.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

TABLE II. Chromatographic conditions

Characteristic	0.46-m tricresyl phosphate column	6.1-m tricresyl phosphate column	0.61-m squalene column
Carrier gas	Helium	Helium	Helium
Flow rate, cc per minute	150 ± 1	55 ± 1	55 ± 1
Temperature, °C	50 ± 1	100 ± 1	185 ± 1
Stationary phase, weight percent	17	17	17
Column diameter, mm	6.35	6.35	6.35
Recorder range, mv	0 to 5	0 to 5	0 to 5
Chart speed, mm per minute	12.7	25.4	50.8

(d) Calibration. Prepare 30 g of a blend of n-butyl mercaptan, isobutyl mercaptan, secondary butyl mercaptan, tertiary butyl mercaptan, and butanol (see 6.4) that approximates the composition of the specimen to be analyzed. Make all weighings to the nearest milligram. Analyze the blend in accordance with the procedure in (e). Determine the area sensitivity of each of the minor components by dividing the areas under the peaks by the respective percent by weight values. Periodically check the calibration with the synthetic blend.

(e) Procedure. Mount the 6.1-m tricresyl phosphate column in the chromatograph and set the temperature at 100° ± 1°C and flow rate at 55 ± 1 cubic centimeters per minute. Allow sufficient time for the instrument to reach equilibrium as indicated by a stable base line on the recorder chart. Set the recorder chart speed at 25.4 mm per minute. Using the conditions specified in table II, inject 15 microliters of the specimen into the column and record the peaks at the sensitivity which allows the maximum peak height. Measure the areas under minor peaks and adjust them to maximum sensitivity. Determine n-butanol in the specimen on the 0.46-m tricresyl phosphate column in a similar manner using the applicable conditions specified in table II. Determine disulfides in the specimen on the 0.61-m squalene column in a similar manner except change the amount of specimen injected to 30 microliters and use the applicable conditions specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

(f) Calculations. Calculate the percent by weight of each minor component as follows:

$$\text{Percent by weight of minor component} = \frac{A}{B}$$

where: A = Square millimeters of area at maximum sensitivity of the minor component as obtained from the applicable chromatogram of the specimen and

B = Square millimeters of area at maximum sensitivity equivalent to 1 percent by weight of the minor component as obtained from the applicable calibration chromatogram in (d).

Calculate the percent by weight of n-butyl mercaptan in the specimen by subtracting the sum of the minor components from 100. Direct percentage reading automatic data processing devices may be used as an alternative to the calculations.

5. PACKAGING

5.1 Interplant shipment (see 6.5). A quantity of n-butyl mercaptan as specified (see 6.2) shall be unit packed and packed to prevent damage in shipment from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation.

5.2 Unitization. When specified (see 6.2), containers shall be palletized in accordance with MIL-STD-147.

5.3 Marking. Shipping containers shall be marked in accordance with Fed. Std. No. 123 and the marking shall include the contract or purchase order number, lot or batch number, date of manufacture, and name and address of the prime contractor. Palletized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. n-Butyl mercaptan is intended for use as a chemical agent simulant.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1).
(c) Palletization, if required (see 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Materials for calibration of assay procedure. The materials used to prepare a blend for calibration in 4.2.4.1(d) may be obtained from Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAC-E, Aberdeen Proving Ground, MD 21010.

6.5 Interplant shipment. Interplant shipment covers packaging for supplies and material which will not enter the military supply system. Typical interplant shipments are from a vendor to a subcontractor or prime contractor or from a vendor or contractor to a military arsenal or plant.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6810-AB15

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-B-51470(EA) N-BUTYL MERCAPTAN, TECHNICAL (METRIC)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSF OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.